

Foundation / ...

Protocol

NSCopying

A protocol that objects adopt to provide functional copies of themselves.

iOS 2.0+ | iPadOS 2.0+ | Mac Catalyst 13.0+ | macOS 10.0+ | tvOS 9.0+ | visionOS 1.0+ | watchOS 2.0+

protocol NSCopying

Overview

The exact meaning of “copy” can vary from class to class, but a copy must be a functionally independent object with values identical to the original at the time the copy was made. A copy produced with [NSCopying](#) is implicitly retained by the sender, who is responsible for releasing it.

[NSCopying](#) declares one method, [copy\(with:\)](#), but copying is commonly invoked with the convenience method [copy\(\)](#). The [copy\(\)](#) method is defined for all objects inheriting from [NSObject](#) and simply invokes [copy\(with:\)](#) with the default zone.

Your options for implementing this protocol are as follows:

- Implement [NSCopying](#) using [alloc](#) and `init...` in classes that don’t inherit [copy\(with:\)](#).
- Implement [NSCopying](#) by invoking the superclass’s [copy\(with:\)](#) when [NSCopying](#) behavior is inherited. If the superclass implementation might use the [NSCopyObject](#) function, make explicit assignments to pointer instance variables for retained objects.
- Implement [NSCopying](#) by retaining the original instead of creating a new copy when the class and its contents are